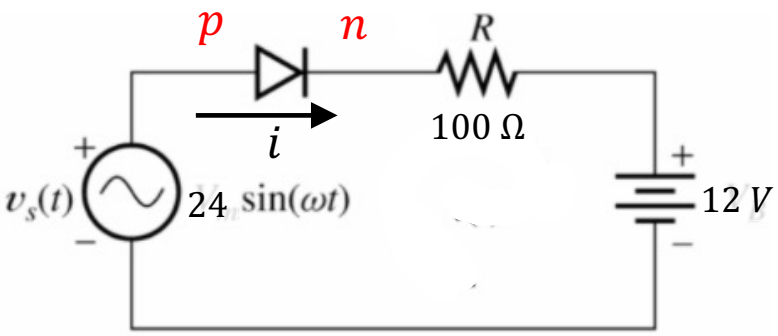
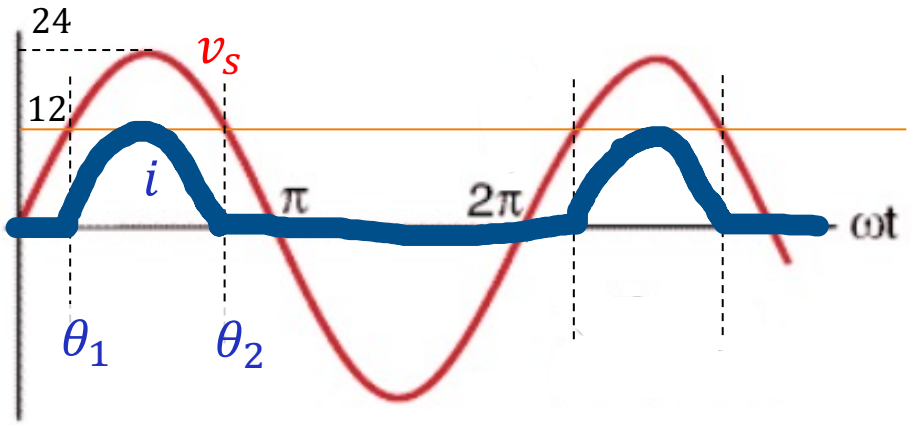
Analysis of Diode Circuits

- Diode Models.
- The ideal model
- The constant-drop model
- The constant-drop-resistance model
- Applications and examples.

Examples

(5): Find the current waveform. Use the ideal diode model





Diode is ON when $v_s \geq 12\ V$

→ Short circuit

$$i(t) = \frac{v_s - 12}{R} = \frac{24 \sin \omega t - 12}{100}$$

$$i_{D(max)} = \frac{24 - 12}{100} = 0.12\ A$$

Otherwise, Diode is OFF

Open circuit, $i = 0$

Conduction angle

$$12 = 24 \sin \theta_1 \quad \Rightarrow \quad \theta_1 = \sin^{-1} \frac{12}{24} = 30°$$

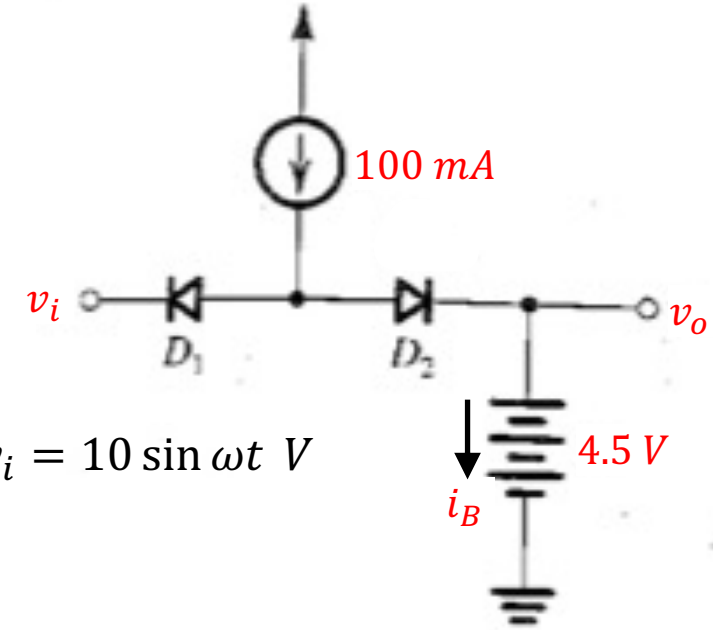
$$\Delta\theta = 180° - 2\theta_1 = 120°$$

Maximum (peak) reverse voltage

$$PIV = 24 + 12 = 36\ V$$

Examples

(6): In the battery-charging circuit, find the average battery current and angle of conduction. Use ideal model.



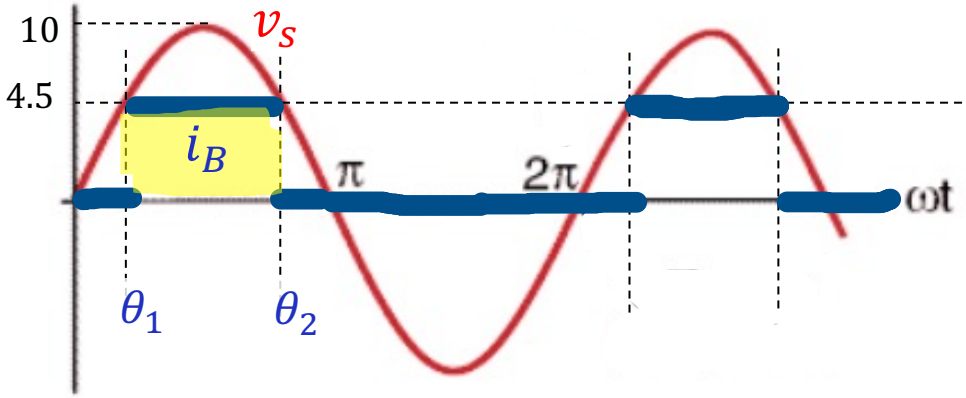
$v_i = 10 \sin \omega t \ V$

$v_i < 4.5\ V$: D1 ON D2 OFF

$i_B = 0$

$v_i > 4.5\ V$: D1 OFF D2 ON

$i_B = 100$ mA



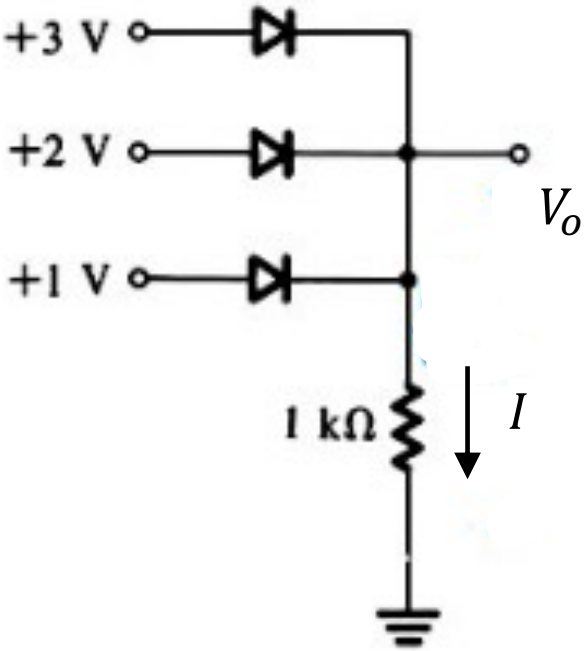
Conduction angle

$\theta_1 = 26.7°$ $\Delta\theta = 180° - 2\theta_1$

Average current?

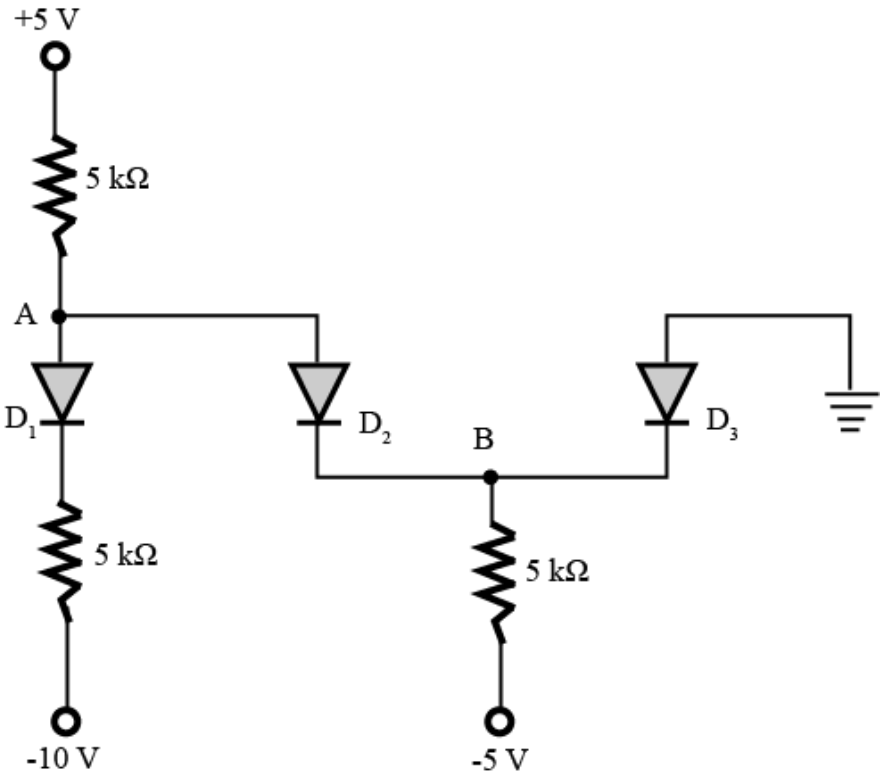
Examples

(7): Analyze the diode circuit below. Use the ideal diode model



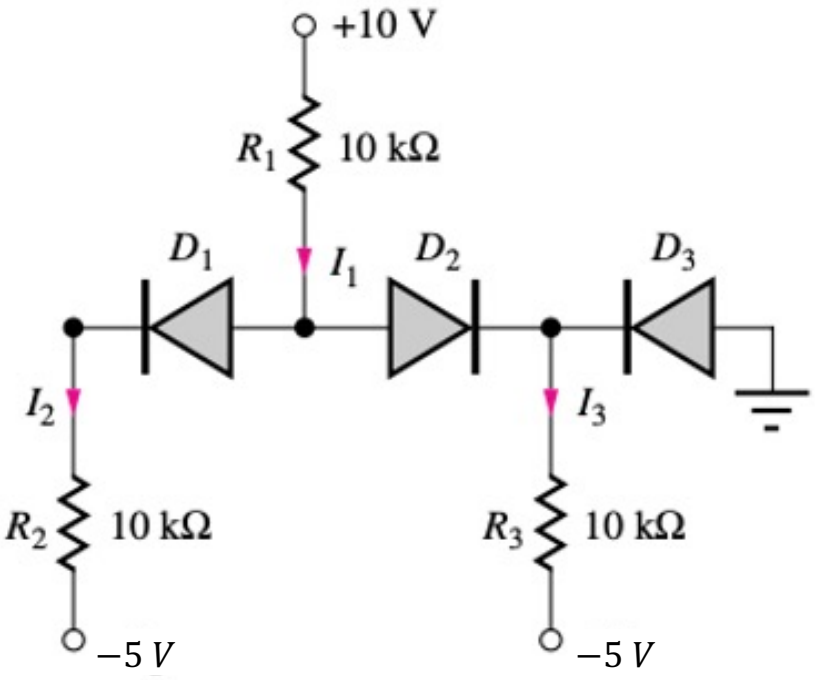
Examples

(8): Analyze the diode circuit below. Use the ideal diode model



Examples

(9): Analyze the diode circuit below. Use the ideal diode model



The Constant-battery Diode Model



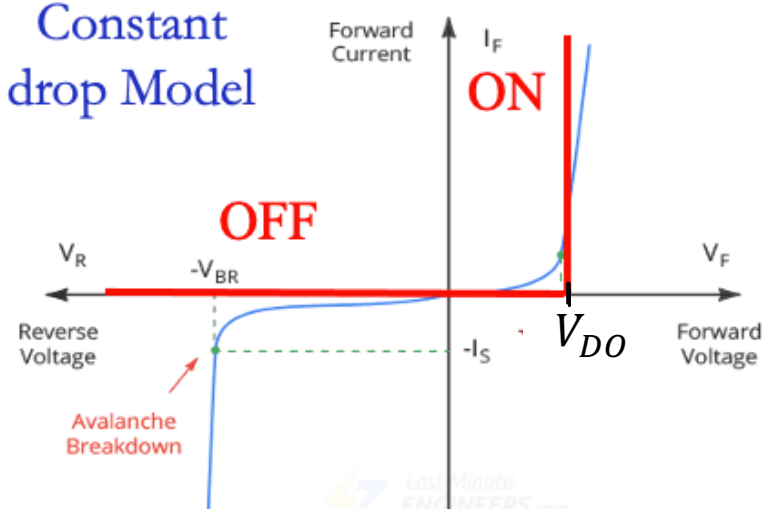
$V_p - V_n \geq V_{DO}$: Diode ON

(Diode is modeled by a constant battery

$I \neq 0, V_D = V_{DO}$)

$V_p - V_n < V_{DO}$: Diode OFF

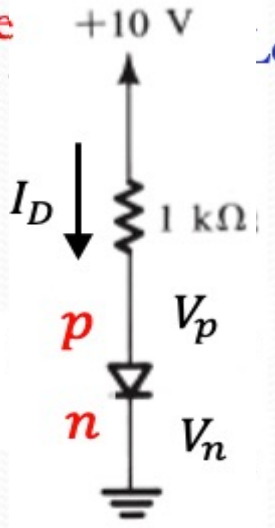
(Diode behaves like an open-circuit $I = 0$)



Advantages: Simple, more accurate

Example

$V_{DO} = 0.7$



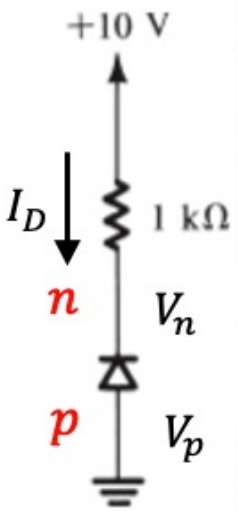
Looks like $V_p - V_n \geq 0.7\ V$

Diode ON

$$I_D = \frac{10 - 0.7}{1k}$$

$$I_D = 9.3\ mA$$

$$V_D = V_p - V_n = 0.7\ V$$



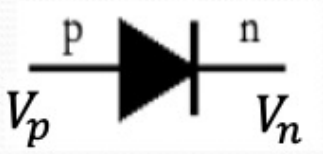
Now $V_p < V_n$

Diode OFF

$$I_D = 0$$

$$V_D = V_p - V_n = -10\ V$$

The Constant-battery-R Diode Model



$V_p - V_n \geq V_{DO}$: Diode ON

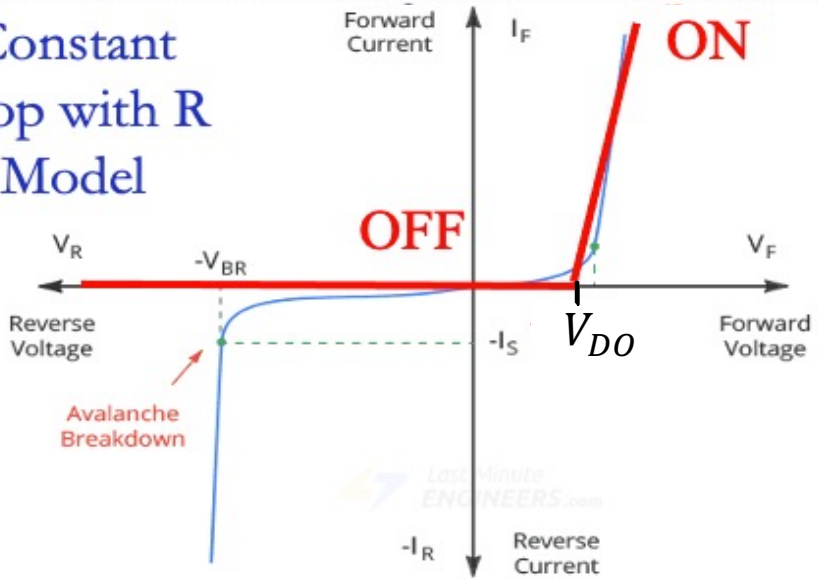
(Diode is modeled by a constant battery-R

$I \neq 0, V_D = V_{DO} + I_D r_D$)

$V_p - V_n < V_{DO}$: Diode OFF

(Diode behaves like an open-circuit $I = 0$)

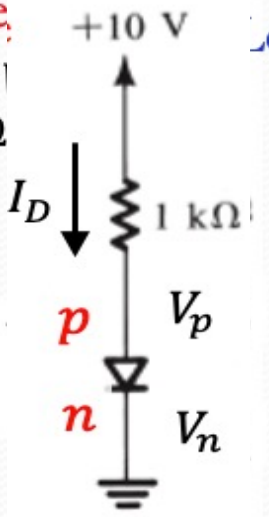
Constant drop with R Model



Example

$V_{DO} = 0.7$

$r_D = 10\Omega$



Looks like $V_p - V_n \geq 0.7\ V$

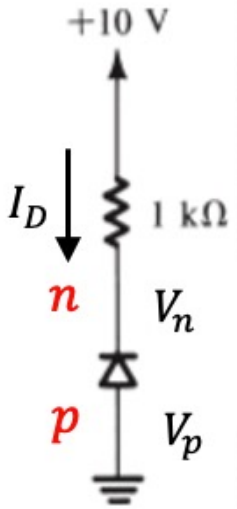
Diode ON

$$I_D = \frac{10 - 0.7}{1k + 10}$$

$$I_D = 9.208\ mA$$

$$V_D = 0.7 + (9.208\ m)(10)$$

$$V_D = 0.792\ \text{V}$$



Now $V_p < V_n$

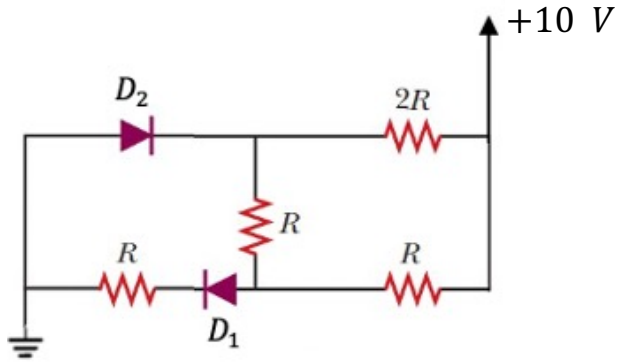
Diode OFF

$$I_D = 0$$

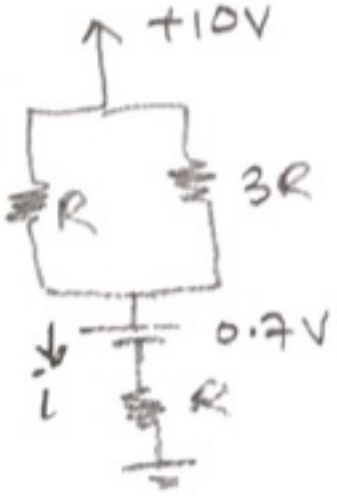
$$V_D = V_p - V_n = -10\ V$$

Examples

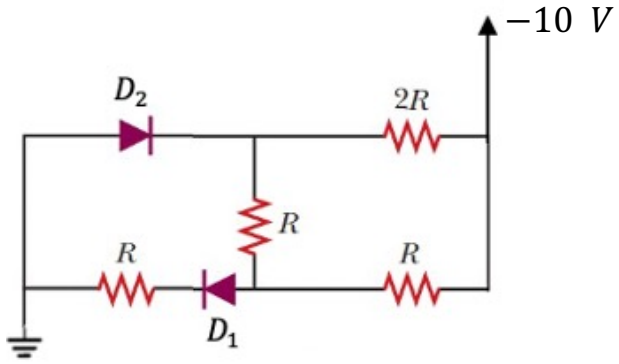
(1): For the given circuit, calculate the current through each diode. For both diodes $V_{DO} = 0.7$ V.



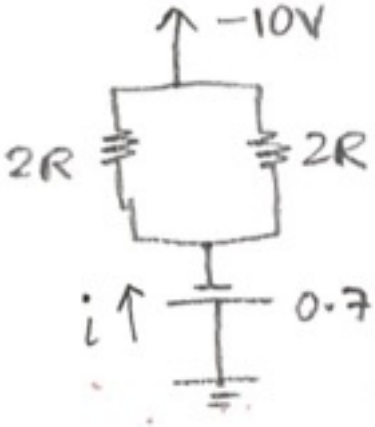
D1 ON D2 OFF



$$i_1 = \frac{10 - 0.7}{(R \parallel 3R) + R}$$



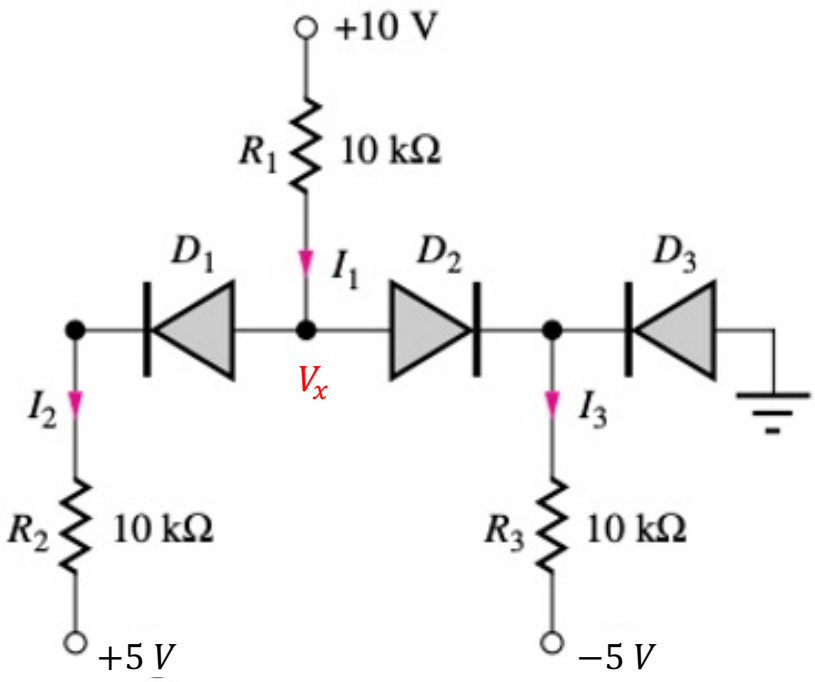
D1 OFF D2 ON



$$i_2 = \frac{10 - 0.7}{(2R \parallel 2R)}$$

Examples

(2): For the given circuit, calculate the current through each diode. For all diodes $V_{DO} = 0.7$ V.



D1 ON D2 ON D3 OFF

$$I_1 = I_2 + I_3$$

$$\frac{10 - V_x}{10} = \frac{V_x - 0.7 - 5}{10} + \frac{V_x - 0.7 + 5}{10}$$

$$V_x = 3.8 \text{ V}$$

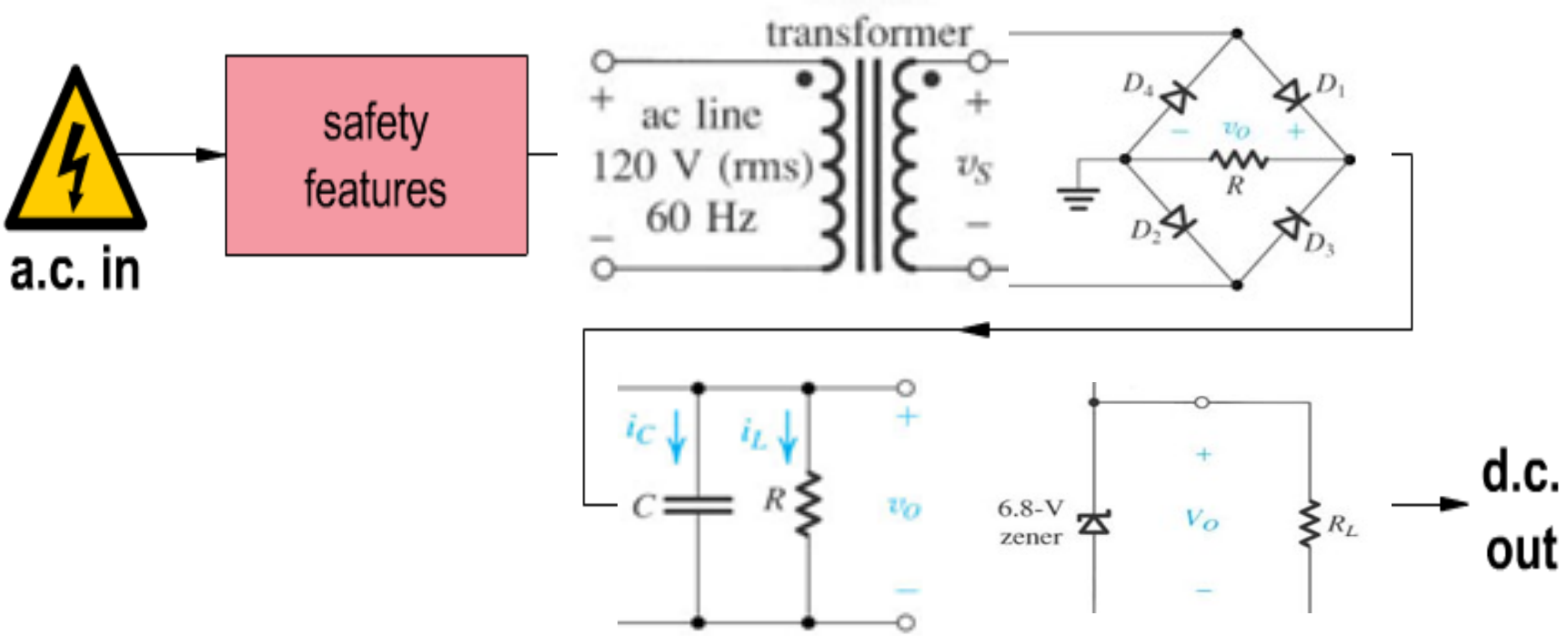
Means that D1 should be OFF

D1 OFF D2 ON D3 OFF

$$I_2 = 0$$

$$I_1 = I_3 = 0.715 \text{ mA}$$

DC Power Supply

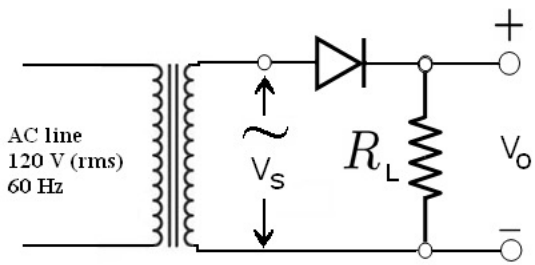


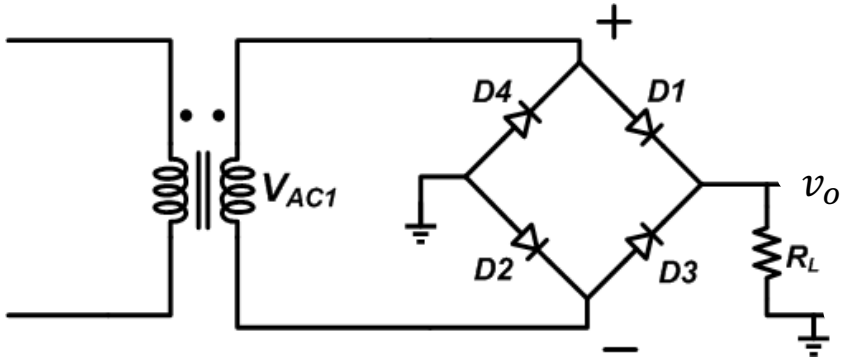


Input | Step-down | Rectifier | Envelop detector | Regulator

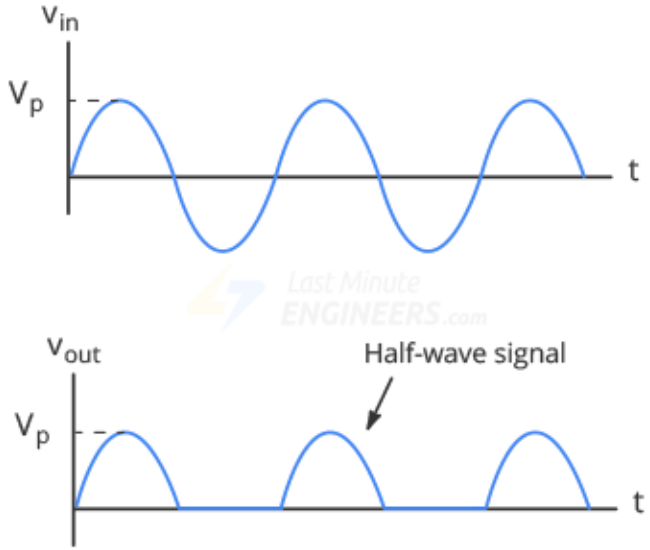
Rectifiers

Rectifiers convert the oscillating input signal to a one-directional signal

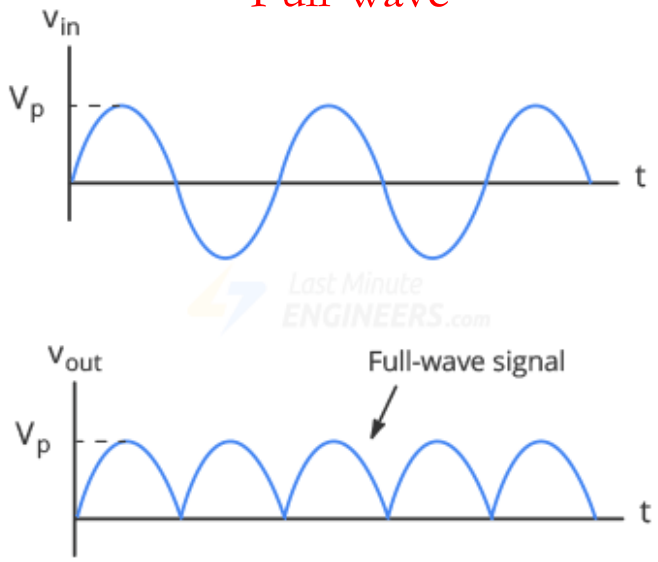




Half-wave

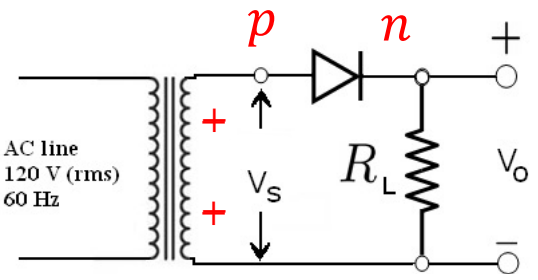


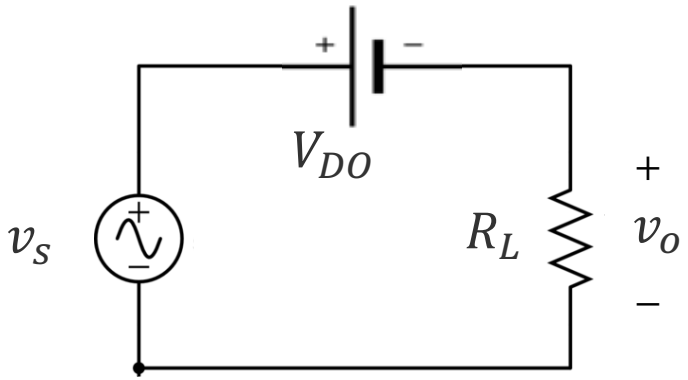
Full-wave



Half-wave Rectifiers

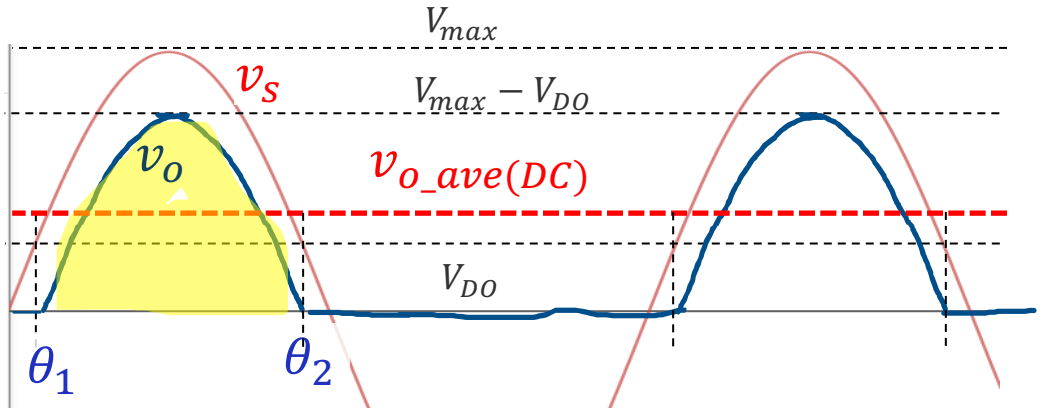
A half-wave rectifier utilizes only half of the input power.





$$v_s(t) = V_{max} \sin \omega t$$

Consider the constant-battery diode model $v_o(t) = v_s(t) - V_{DO}$



$$v_{o_ave} = \frac{1}{2\pi}\int_{\theta_1}^{\theta_2} v_o(\omega t)\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{2\pi}\int_{\theta_1}^{\theta_2} [v_s(\omega t) - V_{DO}]\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{2\pi}\int_{\theta_1}^{\theta_2} [V_{max} \sin \omega t - V_{DO}]\, d(\omega t)$$

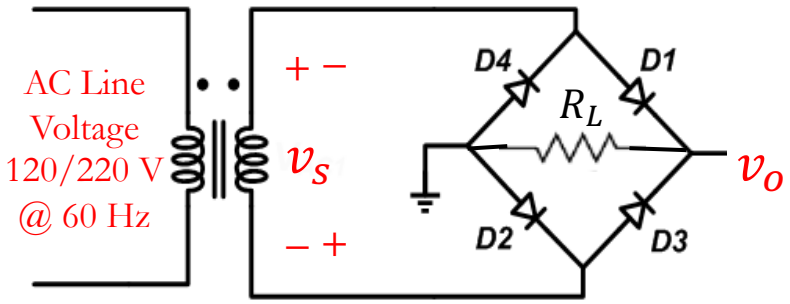
$$v_{o_ave} = \frac{-1}{2\pi}[V_{max}(\cos\theta_2 - \cos\theta_1) + V_{DO}(\theta_2 - \theta_1)]$$

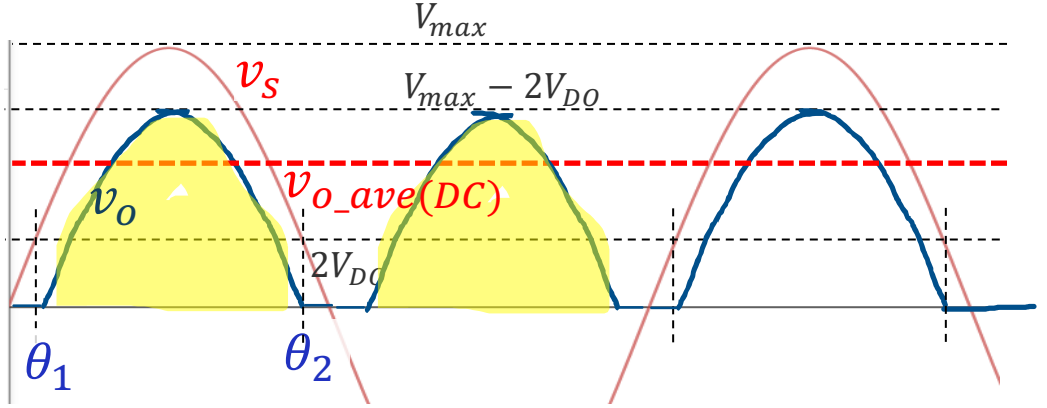
$PIV = V_{max}$ Peak Inverse Voltage

Peak conduction current $I_{max} = \dfrac{V_{max} - V_{DO}}{R}$

Full-wave Rectifiers

A full-wave rectifier utilizes both halves of the input power.





$$v_s(t) = V_{max} \sin \omega t$$

Consider the constant-battery diode model

$$v_o(t) = v_s(t) - 2V_{DO}$$

$$v_{o_ave} = \frac{2}{2\pi} \int_{\theta_1}^{\theta_2} v_o(\omega t)\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{\pi} \int_{\theta_1}^{\theta_2} [v_s(\omega t) - 2V_{DO}]\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{\pi} \int_{\theta_1}^{\theta_2} [V_{max} \sin \omega t - 2V_{DO}]\, d(\omega t)$$

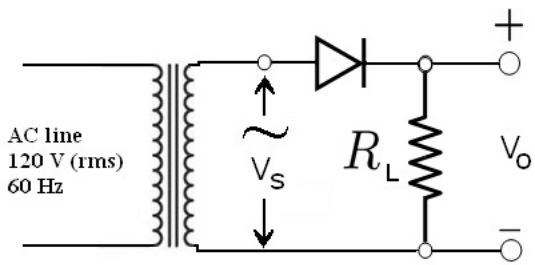
$$v_{o_ave} = \frac{-1}{\pi} [V_{max}(\cos \theta_2 - \cos \theta_1) + 2V_{DO}(\theta_2 - \theta_1)]$$

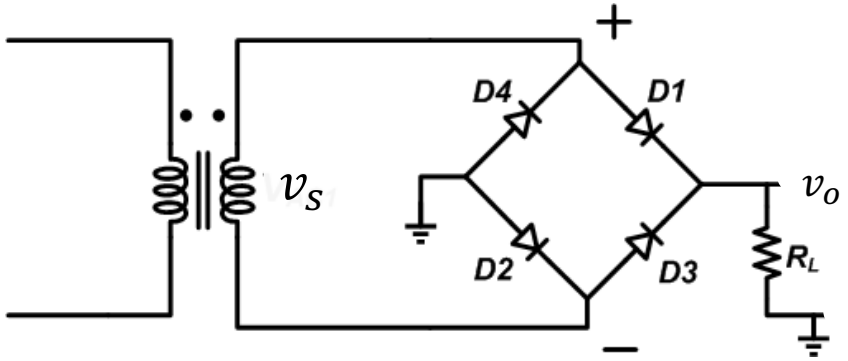
$PIV \approx V_{max}$ Peak Inverse Voltage

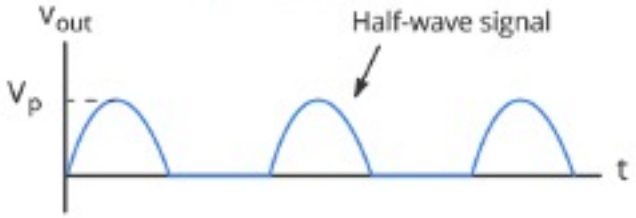
Peak conduction current $I_{max} = \dfrac{V_{max} - 2V_{DO}}{R}$

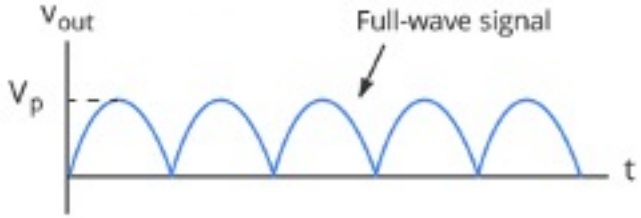
Rectifiers Transfer Function

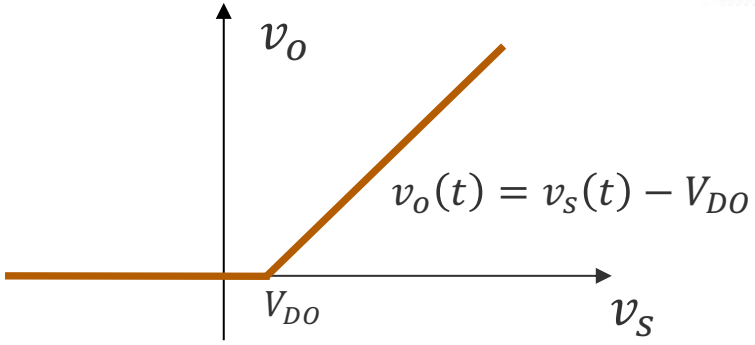
Rectifiers convert the oscillating input signal to a one-directional signal

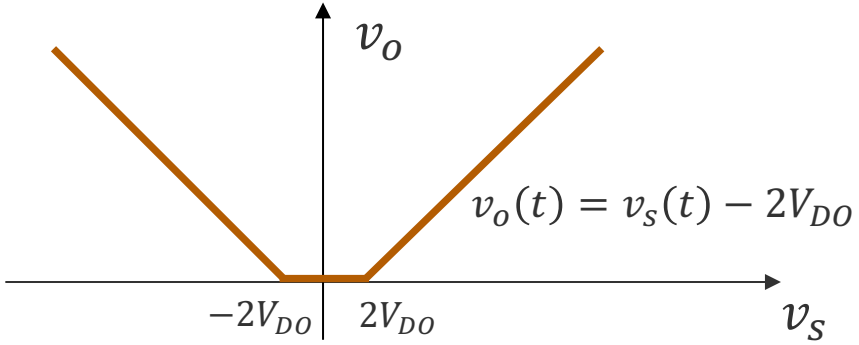






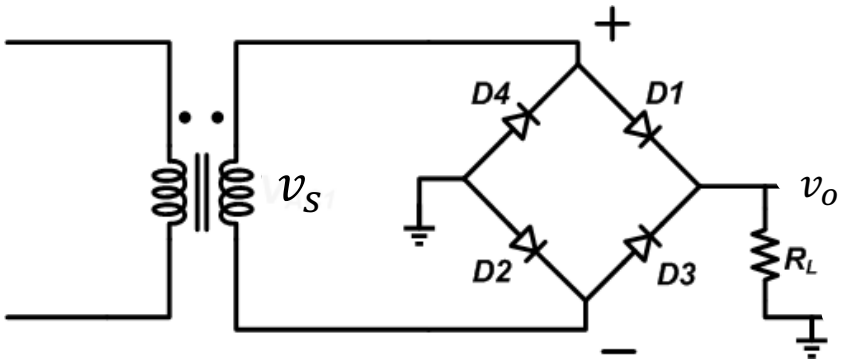






Example

A bridge rectifier is fed from a 120 V (rms) AC source at 60 Hz. The circuit uses a 10:1 transformer with a load of 1 Ωk. Analyze the circuit using the ideal diodes.



$$V_{max} = \frac{120}{10}\sqrt{2} = 12\sqrt{2} \text{ V}$$

$$v_{o_max} = V_{max} - 2V_{DO} = V_{max} = 12\sqrt{2} \text{ V}$$

$$v_{o_ave} = \frac{2V_{max}}{\pi} = 10.8 \text{ V}$$

$$i_{D_ave} = \frac{v_{o_ave}}{R_L} = 10.8 \text{ mA}$$

$$i_{D_max} = \frac{v_{o_max}}{R_L} = 17 \text{ mA}$$

$$PIV = 17 \text{ V}$$

Q & A

Send me an email if
you have any
questions.